

Approximation of Mixed Volumes of Special Convex Bodies

Master's Project Report

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Declaration

This project report is a presentation of my original research work. Wherever contributions of others are involved, every effort is made to indicate this clearly, with due reference to the literature, and acknowledgment of collaborative research and discussions.

Contents

Abstract	2
1 Introduction	3
2 Notations and Basic Setup	5
3 Overview of the steps	6
4 Access to Oracles	7
4.1 Implementation and analysis of Oracle B_1	7
4.2 Implementation and analysis of Oracle B_2	8
4.2.1 Introducing Ω Oracle	8
5 Noisy Binary Search	10
5.1 Behaviour of log concave probability	10
5.2 Main algorithms and its analysis	12
5.3 How fast we can achieve λ using a noisy binary search?	13
5.4 Controlling the error in decision making in algorithm 2	17
6 Algorithm to approximate mixed volumes M_i	18
6.1 Time complexity of the algorithm 3	20
7 Conclusion	21
8 Acknowledgement	22

Abstract

In the area of convex geometry, the approximation of volumes of convex bodies is an active area of research. Recall that the Steiner formula for two convex sets $P, B \in \mathbb{R}^n$ along with a non-negative scalar λ , is given by

$$\text{vol}_n(P + \lambda B) = \sum_{i=0}^n a_i \lambda^i \text{ where } a_i = \binom{n}{i} M_i.$$

It is a polynomial in λ of degree n and the coefficients M_i of λ^i are known as the i -th mixed volume of P, B and often denoted as $M_i = V(P^{(n-i)}, B^{(i)})$. In this paper, we consider two special types of convex bodies. First, a compact convex body $P \subset \mathbb{R}^n$, along with the Euclidean ball $B \subset \mathbb{R}^n$. We propose a method to approximate the n -dimensional measure of mixed volumes M_i . The key idea is to ensure that a random walk, say $\mathcal{W}$ in $P + \lambda B$, should pick points with probability at least $\frac{1}{\text{poly}(n)}$ from the component that corresponds to M_i in $P + \lambda B$.

Chapter 1

Introduction

In the area of convex geometry, we study to the convex sets in $\mathbb{R}^n$. In this thesis, we will talk about a subclass of convex sets in $\mathbb{R}^n$ that are compact and have a non-empty interior, called convex bodies $\mathbb{R}^n$. Let $\mathcal{K}^n$ be the set of all convex bodies in $\mathbb{R}^n$. A linear combination of $K_1, K_2, \dots, K_k \in \mathcal{K}^n$ with coefficients $\lambda_1, \lambda_2, \dots, \lambda_k \geq 0$ is defined as follows

$$\sum_{i=1}^k \lambda_i K_i = \left\{ \sum_{i=1}^k \lambda_i v_i \mid v_i \in K_i \text{ for } i \in \{1, 2, \dots, k\} \right\} \quad (1.1)$$

The eq. (1.1) is known as the weighted Minkowski sum of $K_1, K_2, \dots, K_k \in \mathcal{K}^n$ with respective weights $\lambda_1, \lambda_2, \dots, \lambda_k$. Then an immediate natural question is to ask what we can say about the volume of $\sum_{i=1}^k \lambda_i K_i$ in eq. (1.1). It turns out that the volume of $\sum_{i=1}^k \lambda_i K_i$ is a homogeneous multinomial of degree n in $\lambda_1, \lambda_2, \dots, \lambda_k$ and given by

$$\text{vol}_n\left(\sum_{i=1}^k \lambda_i K_i\right) = \sum_{i_1=1}^k \sum_{i_2=1}^k \cdots \sum_{i_n=1}^k V(K_{i_1}, K_{i_2}, \dots, K_{i_n}) \lambda_{i_1} \lambda_{i_2} \cdots \lambda_{i_n} \quad (1.2)$$

This eq. (1.2) is known as the Steiner-Minkowski formula [Sch13]. the terms $V(K_{i_1}, K_{i_2}, \dots, K_{i_n})$ in eq. (1.2) are called mixed volumes of $K_{i_1}, K_{i_2}, \dots, K_{i_n}$. In the history of convex geometry computing mixed volumes of convex bodies has turned out to be a significantly more difficult problem than approximation of the volume of a single convex body. Only a few results are known so far in this particular area. The key results related to mixed volumes were given by Dyer, Gritzmann, and Hufnagel in [DGH98]. They provided a randomized polynomial-time algorithm to obtain approximations of certain mixed volumes by considering well-presented convex bodies $K_1, K_2, \dots, K_s$. More recently, Narayanan and Roy [NR25] considered a collection of k -convex polytopes $P_1, P_2, \dots, P_k$, each defined as the convex hull of at most m_0 points in $\mathbb{Z}^n$. They gave a randomized polynomial-time algorithm that produces an estimate

within a multiplicative factor of $(1 \pm \epsilon)$ of the true mixed volume with probability at least $1 - \delta$. In this thesis, our goal is to prove the following,

Theorem 1.1 (Main Result:). *Let $P \in R^n$ be a polytope that satisfies*

$$B_n(0, r) \subseteq P \subseteq B_n(0, R) \text{ with } \log(R/r) \leq n^{t_0}.$$

Assume that we have access to :

- *a volume approximation oracle,*
- *a quadratic optimization oracle Ω on P .*

Thus, there exists a randomized algorithm that, for every $k \in \{0, 1, \dots, n\}$ and $\epsilon, \delta > 0$, outputs an estimate $\hat{M}_k$ satisfying

$$\mathbb{P}[(1 - \epsilon)M_k \leq \hat{M}_k \leq (1 + \epsilon)M_k] \geq 1 - \delta$$

in the time polynomial in $n, \log(1/\epsilon), \log(1/\delta), \log(R/r)$.

Chapter 2

Notations and Basic Setup

We have considered the polytope $P \in \mathbb{R}^n$ given by the equation $A_{m \times n} X_{n \times 1} \leq b_{m \times 1}$. Here, the entries of $A_{m \times n} = (a_1, a_2, \dots, a_m)^T$, $a_i \in \mathbb{R}^n$ and $b_{m \times 1}$ are rational entries. Moreover, without loss of generality, one can consider that P contains the origin 0. Also, to ensure that the polytope P is neither too flat nor too large, P must satisfy the following equation

$$B_n(0, r) \subseteq P \subseteq B_n(0, R) \text{ with } \log\left(\frac{R}{r}\right) \leq n^{t_0} \text{ for some fixed } t_0 \in \mathbb{N}. \quad (2.1)$$

The second convex body for us will be a n dimensional euclidean ball B of radius $\tilde{r} > 0$ and the center is the origin 0. The steiner formula eq. (1.2) for two convex bodies $P, B \in \mathbb{R}^n$ along with a non-negative scalar λ , is given by

$$\text{vol}_n(P + \lambda B) = \sum_{i=0}^n \binom{n}{i} M_i \lambda^i. \quad (2.2)$$

It is a polynomial in λ of degree n and the coefficients M_i of λ^i are known as the i -th mixed volume of P, B and often denoted as $M_i = V(P^{(n-i)}, B^{(i)})$ and $a_i = \binom{n}{i} M_i$.

Chapter 3

Overview of the steps

We have the polytope P and ball B , we want to approximate the mixed volume $V(P^{(n-k)}, B^{(k)})$. To make it convenient for the readers, we have given an overview of the steps

1. **Choice of decomposition of $P + \lambda B$:** The decomposition is already known for our case. It is shown in [Sch13], In terms of using normal cones in the facets of the polytope and the appropriate lower dimensional balls.
2. **Scaling of euclidean ball B :** We want to find a scaling factor λ to scale B so that the volume portion contributes $V(P^{(n-k)}, B^{(k)})$.
3. **Algorithm to approximate the mixed volume with the correct choice of λ :** In this part we will describe the algorithm 3 to approximate the mixed volume corresponding to the mixed volume M_i .

Chapter 4

Access to Oracles

In this paper, we will require three types of oracles as a black box for our algorithm.

- **Oracle B_1 :** When querying λ to B_1 , it computes $\sum_{j=0}^n a_j \lambda^j$ approximately, that is, the volume of $P + \lambda B$. The B_1 uses the algorithm based on [Bol].
- **Oracle B_2 :** Querying λ to B_2 , we get a sample i taken from $\{0, 1, 2, \dots, n\}$ with probability $\frac{a_i \lambda^i}{\sum_{j=0}^n a_j \lambda^j}$
- **Oracle B_3 :** Querying λ to B_3 , we get an empirical distribution $\hat{\mathcal{D}}_\lambda$ with a vector $\hat{r} \in \mathbb{R}_{\geq 0}^{n+1}$ with $\sum_{m=0}^n \hat{r}_m = 1$. Thus we will have a sample taken from $\{0, 1, 2, \dots, n\}$ with

$$\mathbb{P}_k(\lambda) = \hat{r}_k + \Lambda \quad (4.1)$$

for each $k \in \{0, 1, \dots, n\}$, $|\Lambda| \leq \frac{1}{n^\gamma}$, $\gamma(\geq 0)$ depending on k and sample size.

4.1 Implementation and analysis of Oracle B_1

We will need to approximate volumes of convex body $P + \lambda B$ where λ is a non-negative scalar. To achieve this goal, we will make an oracle B_1 . There are results in [DFK91] regarding the computation of volumes of convex bodies. Afterwards Bella Bollobas gave a faster algorithm to approximate the volume of convex bodies.

Theorem 4.1 ([Bol], Theorem 4.1). *There is a fully polynomial randomized approximation scheme for the volume of a convex body given by a well-guaranteed membership oracle.*

With the help of this theorem, we can create an oracle B_1 . In this case, we will query λ to B_1 as input. B_1 will be given as an ϵ -approximation of the true volume of $P + \lambda B$. The

size of the input is given by $\iota = n + \langle r + \lambda\tilde{r} \rangle + \langle R + \lambda\tilde{r} \rangle$, where $\langle x \rangle$ represents the number of binary digits of a diadic rational x . From [Sch13], we can say that

$$P + \lambda B = \{x : a_i^T x \leq b_i + \lambda\tilde{r} \|a_i\|_2, i = 1, 2, \dots, m\} \quad (4.2)$$

From eq. (4.2) one can easily get a strong membership oracle for $P + \lambda B$. Consider the following inclusion $B_\infty(0, r/\sqrt{n}) \subset B_n(0, r)$, which is a standard consequence of the inequality $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$; for reference, we have Ball [B⁺97]. It immediately tells us that

$$B_\infty(0, r/\sqrt{n}) \subset B_n(0, r + \lambda\tilde{r}) \subset P + \lambda B \subset B_n(0, R + \lambda\tilde{r}) \subset B_\infty(0, 2(R + \lambda\tilde{r})). \quad (4.3)$$

As [Bol] mentioned, we can take at least $l = O(\log(n))$ concentric cubes C_l , so that $C_0 = B_\infty(0, r/\sqrt{n})$ and $C_l = B_\infty(0, 2(R + \lambda\tilde{r}))$. Defining $K_i = C_i \cap (P + \lambda B)$, we can have

$$\text{vol}_n(P + \lambda B) = \frac{\text{vol}_n(K_l)}{\text{vol}_n(K_{l-1})} \cdot \frac{\text{vol}_n(K_{l-1})}{\text{vol}_n(K_{l-2})} \cdots \frac{\text{vol}_n(K_1)}{\text{vol}_n(K_0)} \cdot \text{vol}_n(K_0) \quad (4.4)$$

note that in eq. (4.4) $K_0 = C_0$ and $K_l = P + \lambda B$. Thus approximation of each $\frac{\text{vol}_n(K_i)}{\text{vol}_n(K_{i-1})}$ will give us an approximation of $\text{vol}_n(P + \lambda B)$. Readers can see [Bol] where bollobas explains it in full detail.

- **Runtime:** If we query an λ to oracle B_1 , then it will give us an ϵ -approximation of $\text{vol}_n(P + \lambda B)$ in $\text{poly}(n, \iota, \log(\frac{1}{\epsilon}))$ times.

4.2 Implementation and analysis of Oracle B_2

Consider the following optimization problem given a point $p \in \mathbb{R}^n$

$$\arg \min_{x \in P} \|x - p\|_2^2 \quad (4.5)$$

Where $P = A_{m \times n} X_{n \times 1} \leq b_{m \times 1}$ with the entries of the matrices A, b being rationals. The one observation is that eq. (4.5), is an optimization problem in the convex set P , but the objective function is quadratic.

4.2.1 Introducing Ω Oracle

We will assume an Oracle Ω that produces $z \in P$ which is an exact optimizer for the convex optimization eq. (4.5) in $O(1)$ times. Before proceeding further, we need restrict ourselves with the following remarks,

- This is an idealized oracle model.
- In practice, this can be implemented approximately via quadratic programming in polynomial time; it is a topic for future work to see whether this approximate oracle suffices for our purposes.

With the help of Ω , we can create an oracle B_2 whose implementation is given below.

Algorithm 1 Query λ as input and oracle Ω for solving quadratic optimization. We want a sample i from $\{0, 1, \dots, n\}$ with probability $\frac{a_i \lambda^i}{\sum_j a_j \lambda^j}$

1. Input λ .
 2. Sample a random point p from $P + \lambda B$.
 3. Consider the optimization problem eq. (4.5). Call Oracle Ω to achieve an v^* optimizer for eq. (4.5).
 4. Take empty set Φ and add a_i to Φ if $a_i^T p = b_i$ holds.
 5. Take a empty matrix U , for each $a \in \Phi$, add a to U as a row. Keep a if $\text{Rank}(U)$ increases, else discard a , until $\text{Rank}(U) = n$ or each $a \in \Phi$ has been checked
 6. $i = \text{Rank}(U)$.
 7. Output i .
-

Chapter 5

Noisy Binary Search

5.1 Behaviour of log concave probability

From eq. (2.2), we have seen $a_i = \binom{n}{i} M_i$. The most interesting property of mixed volumes is that they form a logarithmic concave sequence (log concave sequence) [Sch13][S⁺89], i.e. M_i 's form a log concave sequence, which is defined as

$$M_i^2 \geq M_{i+1}M_{i-1} \implies \log(M_i) \geq \frac{\log(M_{i+1}) + \log(M_{i-1}))}{2}, \forall i \in \{1, \dots, (n-1)\}. \quad (5.1)$$

The binomial coefficient is also logarithmic concave. Thus, the term wise product (Hadamard) of two log concave sequences is also log concave [S⁺89]. Therefore, we have

$$a_i^2 \geq a_{i+1}a_{i-1} \implies \log(a_i) \geq \frac{\log(a_{i+1}) + \log(a_{i-1}))}{2}, \forall i \in \{1, \dots, (n-1)\} \quad (5.2)$$

With some basic algebra, we can observe that

$$p_j^2 \geq p_{j+1}p_{j-1}, \text{ for all } 1 \leq j \leq n-1$$

which is already defined in ???. Let us fix an $t > 11$, this is to indicate an inverse polynomial threshold namely $\frac{1}{n^t}$. Let $P_i(\lambda) = p_i$ be a true distribution corresponding to λ and i . In the case of mixed volumes as λ increases, the probability behavior of the probability distribution really matters. For three different $\lambda_1, \lambda_2, \lambda_3$, with $\lambda_1 < \lambda_2 < \lambda_3$ we have shown a rough picture fig. 5.1 in which along the x-axis we have integers i and along the y-axis, we have $\mathbb{P}_i(\lambda)$. The inverse polynomial threshold $\frac{1}{n^t}$ is the same for all graphs in fig. 5.1 and clearly we observe that as λ increases, the pick of the distribution that lies above the threshold shifts away from 0 and moves towards n after crossing any fixed integer k in between.

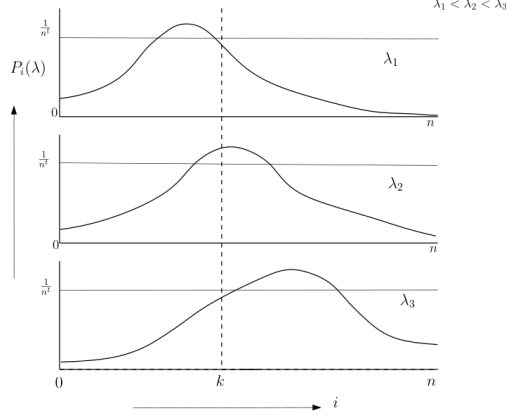


Figure 5.1

This idea motivates us to perform a noisy binary search to locate a λ given k, t for this the pick of the corresponding unknown underline log concave probability distribution lies above the threshold $\frac{1}{n^t}$ at k . This idea also serves us to get a bounded interval in which we can perform the noisy binary search.

From the geometry of $B(0, \tilde{r})$ and eq. (2.1), we have the following $\frac{r}{\tilde{r}}B \subseteq P \subseteq \frac{R}{\tilde{r}}B$ which gives us directly $\frac{\lambda \tilde{r}}{R}P \subseteq \lambda B \subseteq \frac{\lambda \tilde{r}}{r}P$. Let $\mathcal{R}_0$ refer to the following that for $\lambda \leq \mathcal{R}_0$, we will have $\mathbb{P}_0(\lambda) \geq \frac{1}{n^t}$. Thus, to obtain $\mathcal{R}_0$, the following should hold,

$$\frac{\text{vol}_n(P)}{\text{vol}_n(P + \lambda B)} \geq \frac{\text{vol}_n(P)}{\text{vol}_n(P + \frac{\lambda \tilde{r}}{r}P)} \geq \frac{1}{(1 + \frac{\lambda \tilde{r}}{r})^n} \geq \frac{1}{n^t} \quad (5.3)$$

Now we know $(1 + x)^n \leq e^{nx} \implies (1 + x)^{-n} \geq e^{-nx}$. A sufficient condition to hold eq. (5.3) with $x = \frac{\lambda \tilde{r}}{r}$ is given by $e^{-nx} \geq n^{-t} \implies x \leq \frac{t \log(n)}{n}$. Therefore substituting x we get the following,

$$(1 + \frac{\lambda \tilde{r}}{r})^{-n} \geq n^{-t} \iff \lambda \leq \frac{r}{\tilde{r}} \cdot \frac{t}{n} \log(n) = \mathcal{R}_0 \quad (5.4)$$

Alternatively $\mathcal{L}_n$ refers to the following that for $\lambda \geq \mathcal{L}_n$, we will have $\mathbb{P}_n(\lambda) \geq \frac{1}{n^t}$. Thus, to obtain $\mathcal{L}_n$, the following should hold.

$$\frac{\text{vol}_n(\lambda B)}{\text{vol}_n(P + \lambda B)} \geq \frac{\text{vol}_n(\lambda B)}{\text{vol}_n(\frac{R}{\tilde{r}}B + \lambda B)} \geq \frac{\lambda^n}{(\frac{R}{\tilde{r}} + \lambda)^n} \geq \frac{1}{n^t} \quad (5.5)$$

. Let $y = \frac{R}{\tilde{r}\lambda}$, with $\log(1 + y) \leq y$. Now, a sufficient condition is $n \log(1 + y) \leq ny \leq t \log(n)$. Now we have from eq. (5.5)

$$(1 + \frac{R}{\tilde{r}\lambda})^{-n} \geq n^{-t} \iff n \cdot \frac{R}{\tilde{r}\lambda} \leq t \log(n) \iff \lambda \geq \frac{R}{\tilde{r}} \cdot \frac{n}{t \log(n)} = \mathcal{L}_n \quad (5.6)$$

5.2 Main algorithms and its analysis

Theorem 5.1. Choose $t > 1$. Fix $k \in \{0, 1, \dots, n\}$. Then $\exists \lambda \in [\mathcal{R}_0, \mathcal{L}_n]$ so that

$$\mathbb{P}_k(\lambda) = \frac{a_k \lambda^k}{\sum_{i=0}^n a_i \lambda^i} \geq \frac{1}{n^t}$$

for some $t > 11$.

We will prove Theorem 5.1 by giving an algorithm that actually gives the desired result. Our goal is to perform a noisy binary search in $[\mathcal{R}_0, \mathcal{L}_n]$ to obtain such a λ corresponding to given k, t . In the following sections, we will give an algorithm to find a λ that does the job.

Before defining the noisy binary search, we gave some definitions as follows;

- For a particular λ , we will query λ to B_3 to obtain an empirical distribution $\hat{\mathcal{D}}_\lambda$ with a vector $\hat{r} \in \mathbb{R}_{\geq 0}^{n+1}$ with $\sum_{m=1}^{n+1} \hat{r} \cdot \hat{e}_m = 1$
- Define $\mathbb{S}_{\hat{r}} = \{m - 1 | \hat{r} \cdot \hat{e}_m - \frac{1}{n^t} > 0, m \in \{1, 2, \dots, n+1\}\}$
- We will say k is on the left side (respectively, on the right side) of $\hat{\mathcal{D}}_\lambda$ if $k < \min \mathbb{S}_{\hat{r}}$ (respectively, if $k > \max \mathbb{S}_{\hat{r}}$). Thus, if $k \in \mathbb{S}_{\hat{r}}$, we will say that k lies both on the left and on the right of $\hat{\mathcal{D}}_\lambda$.

Algorithm 2 Given $k \in \{0, 1, \dots, n\}$, $t > 11$, and black box B_3 . Perform noisy binary search in $[\log(\mathcal{R}_0), \log(\mathcal{L}_n)]$ to obtain an $\log(\lambda)$, so that $\mathbb{P}_k(\lambda) \geq \frac{1}{n^t}$.

1. $L_0 = \log(\mathcal{R}_0)$, $R_0 = \log(\mathcal{L}_n)$. Take $l = 0$.
 2. $\log(\lambda_l) = \frac{L_l + R_l}{2} \implies \lambda_l = e^{\frac{L_l + R_l}{2}}$.
 3. Query λ_l to B_3 , to get the empirical distribution $\hat{\mathcal{D}}_{\lambda_l}$.
 4. If k lies both on the left and on the right of $\hat{\mathcal{D}}_{\lambda_l}$, it terminates the search and output λ_l , else $l = l + 1$ and goes to the next state.
 5. If k lies on the left side of $\hat{\mathcal{D}}_{\lambda_l}$, update $R_l \leftarrow \log(\lambda_l)$, otherwise if k lies on the right side of $\hat{\mathcal{D}}_{\lambda_l}$, update $L_l \leftarrow \log(\lambda_l)$.
 6. Use updated $[L_l, R_l]$ and continue from state 2.
-

5.3 How fast we can achieve λ using a noisy binary search?

For fixed $t > 11$ and fixed $k \in \{0, 1, \dots, n\}$, we know $\exists$ a true λ_k such that $\mathbb{P}_k(\lambda_k) \geq \frac{1}{n+1}$. Let $I_{k,t}$ be the maximal interval containing $\log(\lambda_k)$ such that for $\log(\lambda) \in I_{k,t}$ we have $\mathbb{P}_k(\lambda) \geq \frac{1}{n^t}$. We would like to show $|I_{k,t}| \geq \frac{1}{e^{\text{poly}(n)}}$. Let us first pay attention to the graph in fig. 5.2

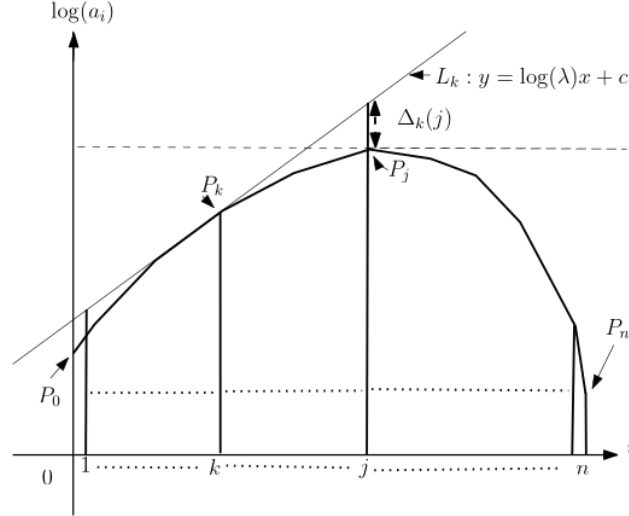


Figure 5.2: graph of $(i, \log(a_i))$ is concave.

considering the integer entries i along the horizontal axes and $\log(a_i)$ along the vertical axes along with the points $P_i \equiv (i, \log(a_i))$ given by the system of equations.

$$L_i : Y = \log\left(\frac{a_i}{a_{i-1}}\right)(X - i) + \log(a_i) \quad (5.7)$$

for each $i \in \{1, \dots, (n-1), n\}$. Let us observe some facts about this graph,

Fact 1: The graph is concave, because eq. (5.2) holds.

Fact 2: We are taking L_i as the tangent at P_i for each $i \in \{1, 2, \dots, n\}$.

Before making progress to find an λ for k , we would require the following lemma,

Lemma 5.2. *Let $y = (\log(\lambda_k))x + c$ be any line that is tangent L_k to the curve given by the system eq. (5.7) at P_k and $\Delta_k(j)$ be the vertical distance from P_j to the tangent L_k for each $j \in \{0, 1, \dots, n\}$. If $|\Delta_k(j)| \leq (t-2) \log(n)$, then $\mathbb{P}_j(\lambda_k) \geq \frac{1}{n^t}$.*

theorem 5.2. The L_k at P_k is given by

$$Y = \log\left(\frac{a_k}{a_{k-1}}\right)(X - k) + \log(a_k).$$

Thus for any integer $0 \leq j \leq n$, $|\Delta_k(j)| = \left| \log\left(\frac{a_k}{a_{k-1}}\right)(j - k) + \log\left(\frac{a_k}{a_j}\right) \right|$. Now for

$$\left| \log\left(\frac{a_k}{a_j}\right) \right| \leq |\Delta_k(j)| \leq (t-2) \log(n) \implies a_j \geq \frac{a_k}{n^{(t-2)}} \implies \mathbb{P}_j(\lambda_k) \geq \mathbb{P}_k(\lambda_k) \cdot \frac{\lambda^{j-k}}{n^{(t-2)}}$$

Since we have assumed λ_k in such a way that $\mathbb{P}_k(\lambda_k)$ is maximum probability among other $\mathbb{P}_j(\lambda_k)$ for $j \neq k$. Thus, $\mathbb{P}_k(\lambda_k) \geq \frac{1}{n+1}$. $\lambda_k \in [\mathcal{R}_0, \mathcal{L}_n]$ and for large n , $(n+1) \approx n$. Therefore we eventually get for large enough n . $\frac{\lambda_k^{j-k}}{n^{(t-2)+1}} \geq \frac{1}{n^t}$. □

Now *theorem 5.2* gives us the hint to find an λ' for k . Observe under similar criterion if we change the role of P_k and P_j and use the tangent $L_j : Y = \log(\lambda') + c'$ at P_j then we can get $\mathbb{P}_k(\lambda') \geq \frac{1}{n^t}$ for some $\lambda' > 0$. Thus, using the noisy binary search, we would try to find integers j_1, j_2 with $j_1 \in (0, k)$, $j_2 \in (k, n)$ and respective $\log(\lambda_1), \log(\lambda_2)$ so that $\Delta_{j_1}(k), \Delta_{j_2}(k) \leq \alpha \log(n)$ where α is a universal constant and $\mathbb{P}_k(\lambda_i) \geq \frac{1}{n^t}$ for $i = 1, 2$. Now, to make the calculation easier, we will consider the following linear transformation T ,

$$T(L_i) : Y = \log\left(\frac{a_i}{a_{i-1}}\right)(X - i) + \log(a_i) - \log\left(\frac{a_k}{a_{k-1}}\right)(X - k) \quad (5.8)$$

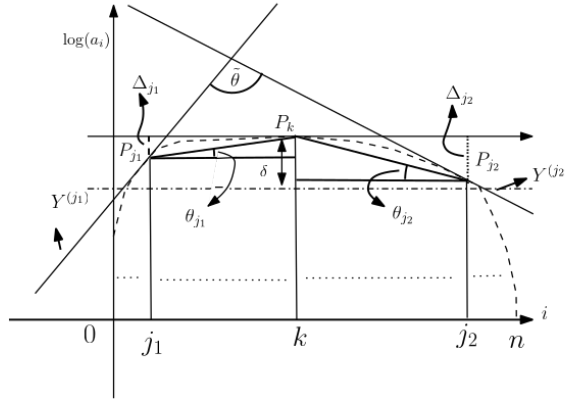


Figure 5.3: Transformation by subtracting line $Y^{(k)}$

The *eq. (5.8)* makes $T(P_k) \equiv (k, \log(a_k))$ the pick point for the transformed graph *fig. 5.3* but keeps it logarithmic concave. The coordinates for each $T(P_i)$'s are $(i, \log(a_i) + \log(\frac{a_k}{a_{k-1}})(k - i))$. For each $i \in \{0, 1, \dots, n\}$, $|\Delta_k(i)|$ will be invariant under transformation *eq. (5.8)*. From *figure fig. 5.3*, $\Delta_k(i) = \log(a_k) - \log(a_i) - \log(\frac{a_k}{a_{k-1}})(k - i) = \log(\frac{a_k}{a_i}) + \log(\frac{a_{k-1}}{a_k})(k - i)$.

Now for any $j \in \{1, 2, \dots, n-1\}$, we want to get a bound for $\Delta_j(0), \Delta_j(n)$ that is the respective vertical distance from $T(P_0), T(P_n)$ to the tangent $T(L_j)$ at $T(P_j)$.

Lemma 5.3. *Given $\log(\frac{R}{r}) \leq n^{t_0}$, we have $|\Delta_j(0)| \leq 2n^{t_0+1}, |\Delta_j(n)| \leq 2n^{t_0+1}$, for any $j \in \{1, 2, \dots, n-1\}$.*

theorem 5.3. First, observe $\Delta_j(0) = \log(\frac{a_j}{a_0}) - j \cdot \log(\frac{a_j}{a_{j-1}})$. Now we can have

$$|\Delta_0| = \left| \log\left(\frac{a_j}{a_0}\right) - j \cdot \log\left(\frac{a_j}{a_{j-1}}\right) \right| \leq \left| \log\left(\frac{a_j}{a_0}\right) \right| + n \left| \log\left(\frac{a_j}{a_{j-1}}\right) \right|$$

$$\Rightarrow |\Delta_0| \leq \sum_{i=1}^j \left| \log\left(\frac{a_i}{a_{i-1}}\right) \right| + n \left| \log\left(\frac{a_j}{a_{j-1}}\right) \right| \leq jn^{t_0} + n^{t_0+1} \leq 2n^{t_0+1}$$

. Here we assumed $\log(\frac{R}{r}) \leq n^{t_0}$. Alternatively, $\Delta_j(n) = \log(\frac{a_j}{a_n}) - \log(\frac{a_j}{a_{j-1}})(j-n)$. Now we can have

$$|\Delta_n| = \left| \log\left(\frac{a_j}{a_n}\right) - \log\left(\frac{a_j}{a_{j-1}}\right)(j-n) \right| \leq \left| \log\left(\frac{a_j}{a_n}\right) \right| + n \left| \log\left(\frac{a_j}{a_{j-1}}\right) \right|$$

$$\Rightarrow |\Delta_n| \leq \sum_{i=j+1}^n \left| \log\left(\frac{a_i}{a_{i-1}}\right) \right| + n \left| \log\left(\frac{a_j}{a_{j-1}}\right) \right| \leq (n-j-1)n^{t_0} + n^{t_0+1} \leq 2n^{t_0+1}.$$

Here we assumed $\log(\frac{R}{r}) \leq n^{t_0}$.

□

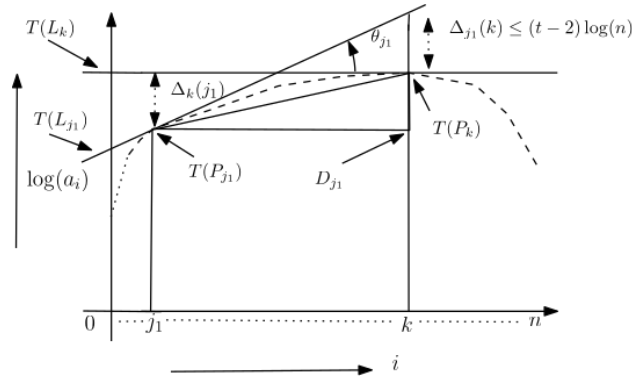


Figure 5.4: graph of $(i, \log(a_i))$ is concave.

Now from *fig. 5.4*, we observe that given the vertical line “ $X = \log(a_k)$ ”, we considered the perpendiculars on it from the points $T(P_i)$ for each $i \neq k$ and called it by D_i . Taking $\Delta T(P_k)T(P_i)D_i$ for each $i \neq k$, we get

$$\tan(\angle T(P_k)T(P_i)D_i) = \left| \frac{\Delta_k(i)}{k-i} \right| \quad (5.9)$$

. For each $i \in \{1, \dots, k\}$, we define $\tan(\theta_{i-1})$ as the respective slope of the lines given by $T(L_i)$ at $T(P_{i-1})$. Alternatively, for each $i \in \{k+1, \dots, n-1, n\}$, we define $\tan(-\theta_i)$ as the respective slopes of the lines given by $T(L_i)$ at $T(P_i)$. Thus, because of the log concave nature, we have the following.

$$\frac{\pi}{2} \geq \theta_0 \geq \theta_1 \geq \dots \geq \theta_{k-1} \geq 0 \geq -\theta_{k+1} \geq \dots \geq -\theta_{n-1} \geq -\theta_n \geq -\frac{\pi}{2}.$$

Now, our analysis will be performed on the basis of the subsequent lemmas.

Lemma 5.4. *Given fixed $k \in \{1, 2, \dots, n-1\}$ and fixed $t > 11$, if we choose integers j_1, j_2 with $j_1 \in (0, k), j_2 \in (k, n)$ such that $|\theta_{j_1}|, |\theta_{j_2}| \leq \frac{(t-2)\log(n)}{\theta(n)}$, then $\Delta_{j_1}(k), \Delta_{j_2}(k) \leq (t-2)\log(n)$.*

theorem 5.4. • Take $j_1 \in (0, k)$, such that $|\theta_{j_1}| \leq \frac{(t-2)\log(n)}{\theta(n)}$.

- From the triangle from by lines $T(P_k), T(P_{j_1})$ and “ $x = \log(a_k)$ ”, we get

$$\frac{\Delta_{j_1}(k)}{|k - j_1|} = \tan(\theta_{j_1}) \approx \theta_{j_1} \leq \frac{(t-2)\log(n)}{\theta(n)}$$

$$\implies \Delta_{j_1}(k) \leq (t-2)\log(n)$$

- Similarly one can show if for an integer $j_2 \in (k, n)$ with $|\theta_{j_2}| \leq \frac{(t-2)\log(n)}{\theta(n)} \implies \Delta_{j_2}(k) \leq (t-2)\log(n)$.

□

Theorem 5.5. *Given fixed $k \in \{1, 2, \dots, n-1\}$ and fixed $t > 11$, we have $|\mathcal{I}_{k,t}| \geq \frac{1}{e^{\text{poly}(n)}}$.*

Theorem 5.5. From theorem 5.4, we have that $|\log(\lambda')| = |\tan(\theta_{j_1})| \approx |\theta_{j_1}| \leq \frac{(t-2)\log(n)}{\Omega(n)}$. Now for any $\log(\lambda) \in [0, \log(\lambda')]$, hence we can conclude that

$$\mathbb{P}_k(\lambda) \geq \frac{1}{n^t}$$

. Therefore $\log(\lambda) \in I_{k,t}$.

$$|I_{k,t}| \supseteq |[0, |\log(\lambda')|]| = \frac{(t-2)\log(n)}{\theta(n)} \geq \frac{1}{\exp(\text{poly}(n))}.$$

□

5.4 Controlling the error in decision making in algorithm 2

In the previous section, we have seen that the noisy binary search will definitely end in $\text{poly}(n)$ steps. Let us observe one step in the noisy binary search. Let that step be w with the interval $[\log(r_{w-1}), \log(l_{w-1})]$. with the mid point $x_w = \frac{1}{2}(\log(r_{w-1}) + \log(l_{w-1}))$. Let μ_w be the log-concave distribution on $\{0, 1, \dots, n\}$ with true mean $p(w)$. and let $\hat{p}(w)$ be the empirical mean. Now, as a decision step, we will go right or left, as was mentioned in steps 4 and 5 of algorithm 2.

We show that $\mathcal{I}_{k,t}$ is not too small. That implies that all the intervals containing $\mathcal{I}_{k,t}$ are at least $\frac{1}{\exp(\text{poly}(n))}$ large. Using hoeffding, we have the following

$$\begin{aligned} \mathbb{P}[|\hat{p}_w - p_w| \geq \epsilon] &\leq 2 \exp(-2N\epsilon^2 n^{-2}) \\ \implies \mathbb{P}[X_w = 1] &\geq 1 - 2 \exp(-2N\epsilon^2 n^{-2}) = \gamma \text{ (say)} \end{aligned}$$

Here, X_w is defined as follows:

$$X_w = \begin{cases} 1 & \text{if the direction chosen at step } w \text{ is correct} \\ 0 & \text{otherwise} \end{cases}$$

In this manner, we have variables $X_1, X_2, \dots, X_T$ where $T = \text{poly}(n)$. We will use a union bound to get the following probability

$$\mathbb{P}[\exists w : X_w = 0] \leq \sum_{w=1}^T \mathbb{P}[X_w = 0] \leq T(1 - \gamma) \quad (5.10)$$

Therefore, the probability that every choice of steps in NBS is correct is given

$$\mathbb{P}[X_1 = X_2 = \dots = X_T = 1] = 1 - \mathbb{P}[\exists w : X_w = 0] \geq 1 - T(1 - \gamma) \quad (5.11)$$

Therefore, substituting γ into eq. (5.11), we get

$$\mathbb{P}[X_1 = X_2 = \dots = X_T = 1] \geq 1 - 2Te^{(-2N\epsilon^2 n^{-2})} \quad (5.12)$$

Therefore, from eq. (5.12), to ensure

$$\mathbb{P}[X_1 = X_2 = \dots = X_T = 1] \geq 1 - \delta \text{ when } \delta \leq 2Te^{(-2N\epsilon^2 n^{-2})} \implies N = \text{poly}(n, \log(1/\delta))$$

Chapter 6

Algorithm to approximate mixed volumes M_i

Algorithm 3 Given a fixed $k \in \{0, 1, 2, \dots, n\}$

1. If $k = 0$, then assign $\lambda = 0$ and query this λ to oracle B_1 , to get an ϵ -approximation of $M_0 = \text{vol}_n(P)$ and output the estimate.
2. If $k = n$, then assign $M_n = \text{vol}_n(B(0, \tilde{r}))$ and output M_n .
3. If $k \in \{1, 2, \dots, n-1\}$, choose an $t > 11$. Using this calculate $\mathcal{R}_0$ and $\mathcal{L}_n$ using eq. (5.4) and eq. (5.6) respectively.
4. Implement algorithm 2 to get an $\log(\lambda^*) \in [\log(\mathcal{R}_0), \log(\mathcal{L}_n)]$ for which $\mathbb{P}_k(\lambda) \geq \frac{1}{n^t}$.
5. Query this λ^* to oracle B_1 to get an ϵ -approximation of $\text{vol}_n(P + \lambda^*B)$. Call the estimate by $V^*(P + \lambda^*B)$.
6. Again query this λ^* to oracle B_3 to get an ϵ -close empirical distribution $\hat{p}_\lambda$ with a vector $\hat{r} \in \mathbb{R}_{\geq 0}^{n+1}$ with $\sum_{m=1}^{n+1} \hat{r} \cdot e_m = 1$ satisfying the following $\{0, 1, 2, \dots, n\}$ with

$$p_k(\lambda) = \mathbb{P}_k(\lambda) = \hat{r} \cdot e_{k+1} + \Lambda$$

for each $k \in \{0, 1, \dots, n\}$, $|\Lambda| \leq \frac{1}{n^\gamma}$, $\gamma(> 0)$ depending on k and sample size.

7. Compute $\hat{M}_k = \hat{r} \cdot \hat{e}_{k+1} \times \frac{V^*(P + \lambda^*B)}{\binom{n}{k} \lambda^{*k}}$
 8. Output the approximate k -th mixed volume $V(P^{(n-k)}, B^{(k)}) = \hat{M}_k$.
-

6.1 Time complexity of the algorithm 3

The time complexity of algorithm 3 is given as follows

1. Steps 1 , 2 and 3 take $O(1)$ times.
2. In step 4 algorithm 2 takes $poly(n)$ times.
3. In step 5 we consume for the oracle B_1 , $poly(n, \iota, \log(\frac{1}{\epsilon}))$ times.
4. In step 6, we need $O(N)$ times to get an empirical estimate.
5. Steps 6 and 7 again take at most $O(1)$ times.

Thus, the total time complexity is given by $poly(n, \iota, \log(1/\epsilon)) + O(N)$.

Chapter 7

Conclusion

- We have designed a randomized polynomial time algorithm to approximate mixed volumes $V(P^{(n-\alpha)}, B^{(\alpha)})$.
- Here we assumed that there is an oracle which gives us a perfect estimator and perfect optimizer for a quadratic optimization problem over a convex set.
- I strongly believe this work can be extended for any strongly convex body instead of an euclidean ball. It can be a potential research problem for future work.

Chapter 8

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